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idity and mortality rates.^{1,2}

e the most common among fractures
America, around 300,000 fractures
zil around 50,000 per year.¹ Tibial

fractures usually occur in two forms: spiral fractures, which are more frequent in people aged over 50 years and result from low-energy trauma, and transverse and/or comminuted fractures, which are more related to high-energy trauma in people aged over 30 years. Low-energy trauma is more related to falls from one's height and sports injuries, while high-energy trauma is more associated with automobile accidents.²

The most used classification is the Orthopaedic Trauma Association/Arbeitsgemeinschaft für Osteosynthesefragen (OTA/AO), which considers the mechanism, location and energy of the trauma. Type "A" stands for simple fractures, type "B" for fragmented fractures with wedges and type "C" for complex multifragmented fractures.^{3,4} Due to force transmission mechanisms and local anatomical characteristics, diaphyseal tibial fractures can have some complications and association with other injuries, such as compartment syndrome, ankle injuries, extension of injury to the tibial plateau, knee ligament injury, among others, which can trigger greater functional impairments and risks.^{1,2}

The most used diagnostic method is radiography. In cases in which there is a suspicion of associated joint and/or ligament injuries, computed tomography or even, in some cases, magnetic resonance imaging is recommended.^{5,6}

Treatment for diaphyseal tibial fractures can be conservative or surgical, with the latter being increasingly used in orthopedic practice. There are different options of surgical treatment,^{1,3,4} which can take into account factors such as fracture type, age, among others, aiming at fracture stability, rehabilitation and an early return to activities, in addition to avoiding postoperative complications.⁴⁻⁶ Since traumatology of the musculoskeletal system comprehends an important part of hospital care and accounts for high rates of morbidity and mortality, especially related to postoperative complications,^{4,6} it is very important to properly understand the current epidemiological situation of this condition in our environment. In Brazil, few epidemiological studies on this topic have been published in recent years, and there is currently no basis for a better understanding of the impact of tibial fractures on the health system and, consequently, for the development of preventive strategies to reduce the risk of comorbidities and complications. Thus, the objectives of this study were to outline the epidemiological profile of ankle fractures treated in a tertiary hospital and to explore associations between fracture characteristics and clinical outcomes of postoperative complications.

MATERIAL AND METHODS

This is a retrospective study conducted by searching the electronic Patient Records System (PRS) for hospital admission records of patients with diaphyseal tibial fractures who were hospitalized and treated between January 2019 and December 2021. Records with incomplete data, patients under 18 years of age or transferred to other hospital services were excluded.

All the information used in this project was collected after the participants signed the Informed Consent Form (ICF) issued by

postoperative complications (infections, loss of synthesis material, wound dehiscence) and death; and secondary, length of hospital stay and surgical treatment variables.

Statistical analysis

Categorical data were expressed as relative frequency (%), continuous data were expressed as mean and standard deviation, and discrete data were expressed as median and quartiles 25-75%. Firstly, a descriptive analysis was conducted to verify the prevalence and distribution of clinical, anthropometric and surgical characteristics of patients.

To explore the association between fracture types "A", "B" or "C" and the clinical outcomes, a generalized linear model was performed with the distributions linear, *gamma* and logistics, according to the nature of the dependent variables. The postoperative complication outcome was determined by the existence of at least one clinical event considered not expected in the postoperative period, such as loosening of surgical material, need for reoperation, surgical site infection, pneumonia, associated fracture and cardiorespiratory arrest. The adopted level of statistical significance was $p < 0.05$. The software used was JAMOVI (Version 1.6.23.0).

RESULTS

The data related to sociodemographic characteristics are shown in Table 1. The study involved 100 patients: 86% of the participants were male and 14% female, with a mean age of 35.8 ± 14.6 years.

Table 1. Sample distribution by gender and age.

Gender	n	%	Mean	M	Ma
Female	14	14.00	45.00	18	72
Male	86	86.00	33.93	18	71
Total	100	100.00	35.82	18	72

Legend: n - number of patients; % - percentage; M - minimum; Ma - maximum

Table 2 presents the data regarding the AO classification of fractures, with the most frequent fractures being the simple type (A), followed by wedge (B) and multifragmented fractures (C). A higher prevalence of simple type (A) fractures was found, with 58% of all fractures.

Table 2. Distribution of diaphyseal tibial fractures as classified by the Orthopaedic Trauma Association / Arbeitsgemeinschaft für Osteosynthesefragen.

Fracture type	N	Percentage (%)
A1	22	22.00
A2	16	16.00
A3	20	20.00
B1	2	2.00
B2	16	16.00
B3	4	4.00
C1	1	1.00
C2	10	10.00

of trauma.

	N	Percentage (%)
Automobile accident	62	62.00
Pedestrian collision	14	14.00
Fall	15	15.00
Sports practice	3	3.00
Could not report	6	6.00

Legend: N - number of patients

The most used treatment was the intramedullary stem (IM), followed by locking plates, conservative treatment and treatment with Ilizarov-type linear or ring external fixators (Table 4).

Table 4. Treatment

Treatment	N	Percentage
External fixator	4	4.00
Intramedullary stem	83	83.00
Plates and screws	8	8.00
Conservative treatment	5	5.00

Legend: N - number of patients

Considering the classification of fractures into types A, B and C, it was found that fracture severity was a determining factor in the outcome of postoperative complications. Patients with a type C fracture were 4.9 times more likely to develop postoperative complications compared to those with a type A fracture (OR = 4.88 [95% CI = 1.61 – 15.46], $p = 0.006$, Table 5). No difference was found when comparing patients who evolved with a type B fracture compared to types A and C.

Table 5. Odds ratio of clinical outcomes by group.

Postoperative complications						
Comparison of groups			OR(B)	CI 95%	Z	P value
C	-	A	4.88	1.61 – 15.46	2.77	0.006*
B	-	A	0.73	0.15 – 2.70	-0.45	0,655

Notes: OR - Odds ratio, CI - Confidence interval

The mean length of hospital stay was 6 (3 - 8.75) days for patients with a type A fracture, 5 (3-7) days for patients with a type B fracture, and 6 (4 - 10.3) days for patients with a type C fracture. There was no association between fracture type and length of hospital stay (Kruskal Wallis = 1.42, $df = 2$, $p = 0.491$). Individuals who developed complications in the postoperative period had a longer

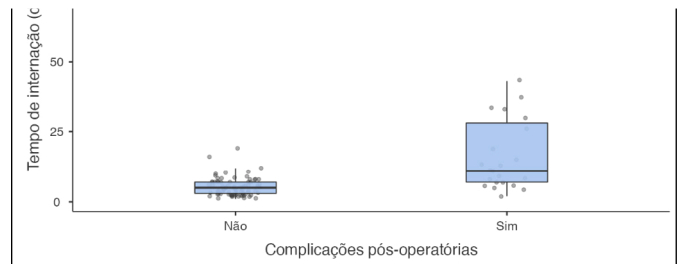


Figure 1 - Relationship between length of stay and complications

DISCUSSION

One hundred patients with diaphyseal tibial fractures were studied, predominantly young adult males. This predominance of males has also been reported in other studies, such as Grecco's⁷, in which 73.7% of the sample was composed of men. The mean age of the patients in this study was 35.82 years (18-72 years), which corroborates other studies found in the literature,^{8,9} such as those by Ali⁸ and Grecco⁷, who found mean ages of 38 and 32 years respectively.

Regarding fracture classification, simple (type A) fractures were the most found type, like the data from other studies that report a higher prevalence of simple fractures, also in hospitalized individuals.^{8,9} The higher incidence in working-age men is directly linked to those previously mentioned factors, especially as regards the mechanism of injury, usually of high energy.¹⁰

The literature notes that there are several methods of treatment for diaphyseal tibial fractures, as well as discussions favoring one method or another. A study conducted by Canadian surgeons showed that 80% of them treated diaphyseal tibial fractures surgically.⁶ Nevertheless, in a conference of the Orthopaedic Trauma Association, 70% of orthopedic surgeons opted for surgical treatment of this injury. Over the years, conservative treatment has become less common due to the risks of non-consolidation and possible displacement of the fracture during treatment.^{5,6}

The intramedullary stem is considered by many surgeons to be the most effective treatment for these fractures.^{6,10} At this point, depending on when the study was conducted, the literature was conflicting. In the study by Grecco et al.,⁷ of the 179 diaphyseal tibial fractures, 86 underwent treatment with cast immobilization, 71 with external fixators and cast immobilization, 17 with plates and screws, and 5 with external fixators only. On the other hand, the study by Vieira Jr. et al.¹⁰ showed a divergent situation, in which of the 123 fractures studied, 37.5% were definitively treated with dynamic compression plates (DCP), 20.3% with bridge plating and intramedullary stems, 15.6% with linear external fixators and 6.3% with Ilizarov-type ring external fixators.¹⁰

The mean length of hospital stay in this study was 8.2 days, a considerably lower number compared to data from other reported authors, who obtained a mean length of hospital stay of 26.61 days.¹¹ The literature indicates that length of hospital stay is directly related to the number of procedures to which a patient is submitted.¹²

As in other studies, simple fractures (42A) were the most found type, while the wedge fractures (42B) and the complex, multifragmented fractures (42C) shown were not similar, but had no statistical significance.¹²

Vieira Junior et al., in a retrospective, descriptive and analytical study

The Ministry of Health warns that automobile accidents are the second largest cause of external deaths in Brazil. In 2017, 35.3 thousand people lost their lives due to a kind of automobile accident. It also reports that approximately BRL 260 million are spent annually

This study reinforces relevant data on diaphyseal tibial fractures with a robust number of participants. It corroborates important aspects regarding the proper care of these injuries, which will allow us to continuously improve treatment and obtain increasingly favorable outcomes.

Diaphyseal tibial fractures had a higher incidence in males aged between 20 and 39 years. The most frequent cause was automobile accidents, mostly causing a simple (type A – AO) fracture, while patients with a type C fracture were more likely (4.9 x) to develop complications. Surgical treatment using an intramedullary stem was the most common method, accounting for 83% of cases, with a mean length of hospital stay of 8.42 days.

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